

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2336

Bayfront 138 kV
July 04, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2336. The Interconnection Customer proposes a 101.1 MW solar plus storage facility interconnecting at the Bayfront 138 kV substation in the Maryland area. The request was received on July 19, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 401 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,751,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cloverleaf Substation-Gunpowder 138 kV Line	Hidalgo Smelter 345/1kV Transformer Bank	116.6	276	29.9
Hazelwood Substation-Western Digital 138 kV	Hidalgo Smelter 345-1 kV TX	101.0	356	12.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gunpowder-Cloverleaf 138 kV line	Rebuild line with higher-capacity conductor	\$6,749,000	78% (\$5,294,000)
Hazelwood Substation-Western Digital 138 kV	Reconductor line	\$686,000	66% (\$457,000)
	Total allocated to project		\$5,751,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.